

Experiment -1

Linux Programming Environment & Basic Commands

- **Setting up of Linux Programming Environment :**

We can perform C programming using either online tools or offline desktop setups depending on convenience.

Online:

- **CoCalc.ai:** Create an account, start a new project, open a Linux terminal window, write your C file using the built-in editor, and run it using terminal commands.

Offline:

- **Dual boot:** Installs Linux alongside Windows on a separate disk partition so you can select which OS to boot when turning on the PC.
- **Single boot:** Replaces the main OS entirely, making Linux the primary operating system on the computer.
- **Bootable Pendrive:** Allows you to boot and run Linux directly from a flash drive without modifying your computer's hard drive.
- **Virtual Machine (VM):** Runs Linux inside host software on your existing operating system without changing your main OS setups.

1. Virtual Machine Manager (Hypervisor)

Type-1 (Baremetal): Runs directly on the computer hardware without needing a host operating system.

a. *Examples:* VMware ESXi, KVM.

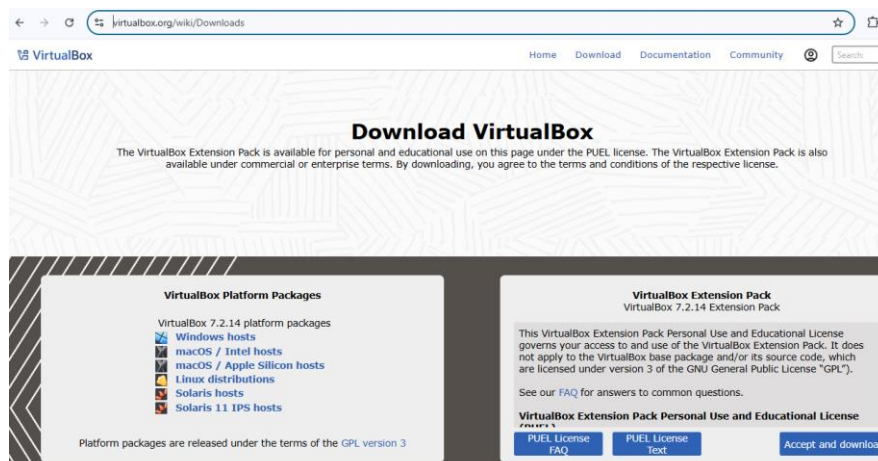
Type-2 (Hosted): Runs as an application inside an existing operating system like Windows.

b. *Examples:* Oracle VirtualBox, VMware Workstation.

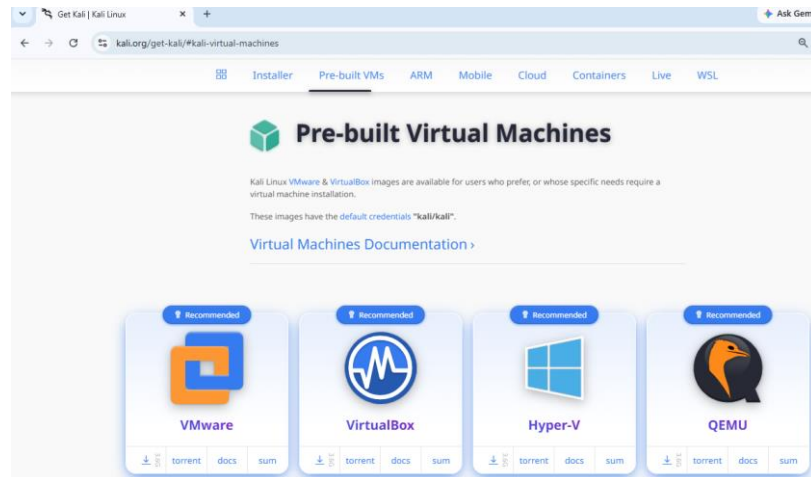
c. *Note:* We are using **Oracle VirtualBox** for this lab.

2. Installation of VMM & VM

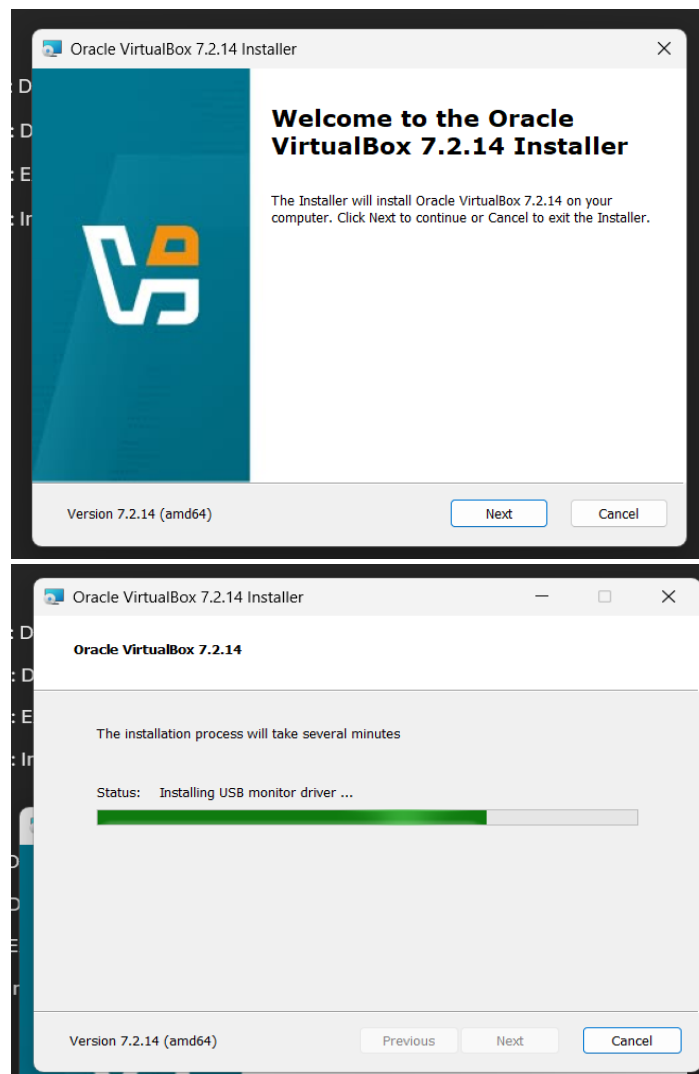
Step-1: Download Oracle VirtualBox setup for Windows from its official site.



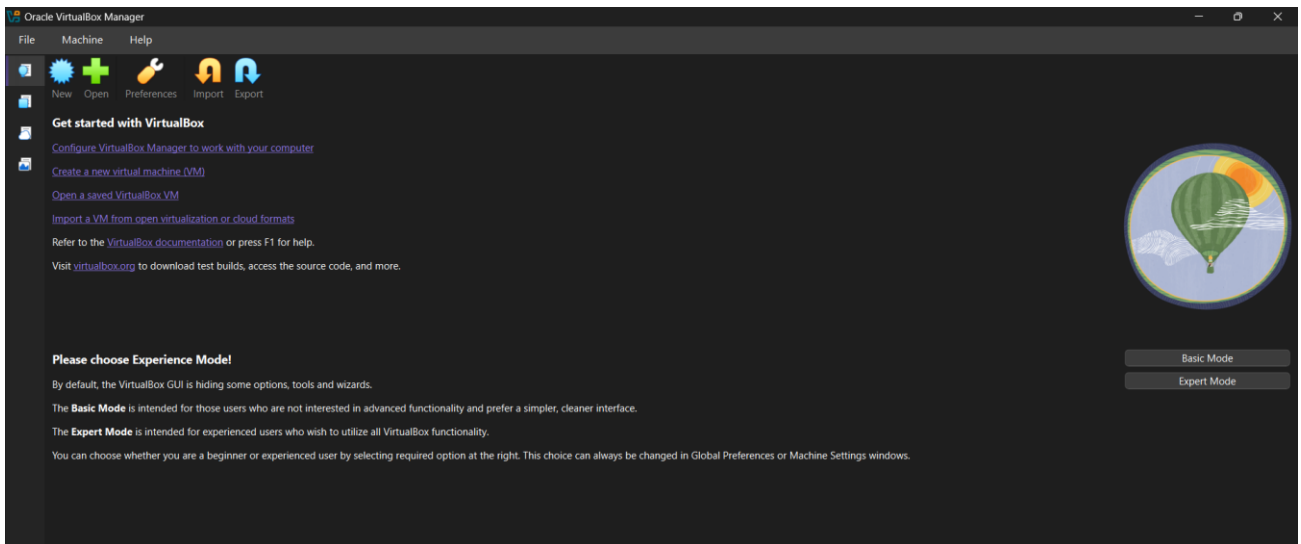
Step-2: Download the pre-built Kali Linux VM image for VirtualBox.



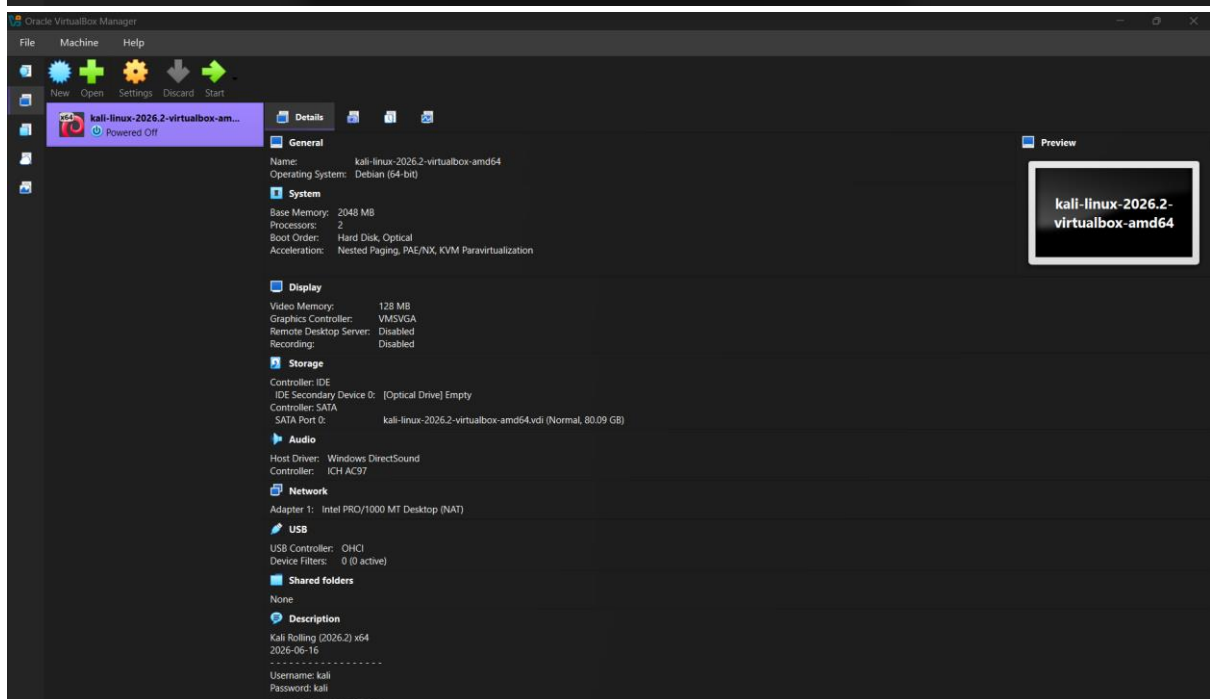
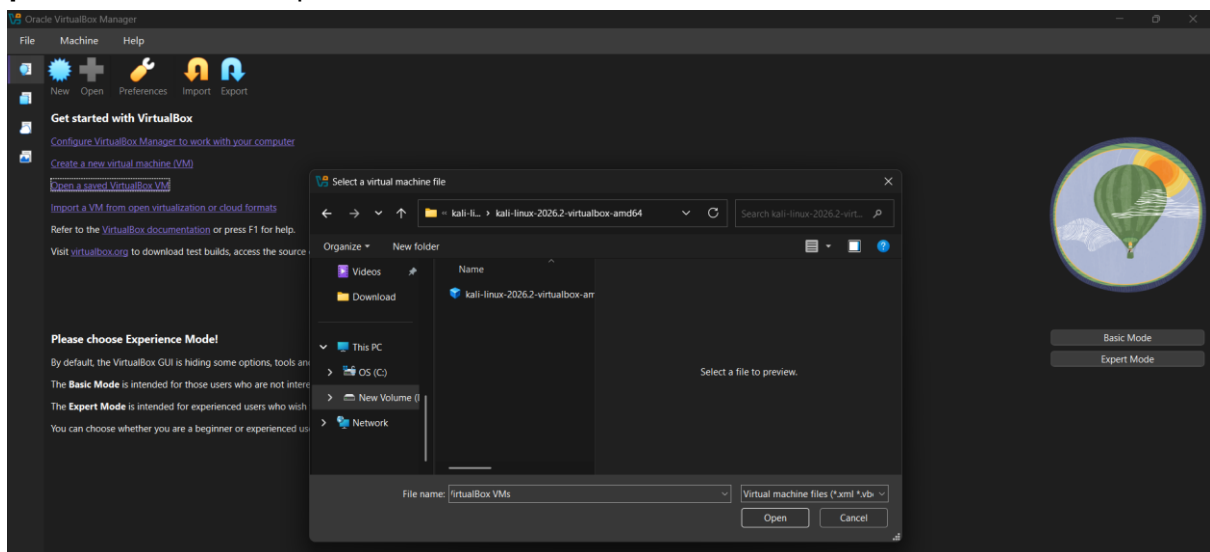
- Step-3 : Extract the downloaded Kali Linux zip folder to a local directory.
- **Step 4:** Run the installer and complete VirtualBox installation.



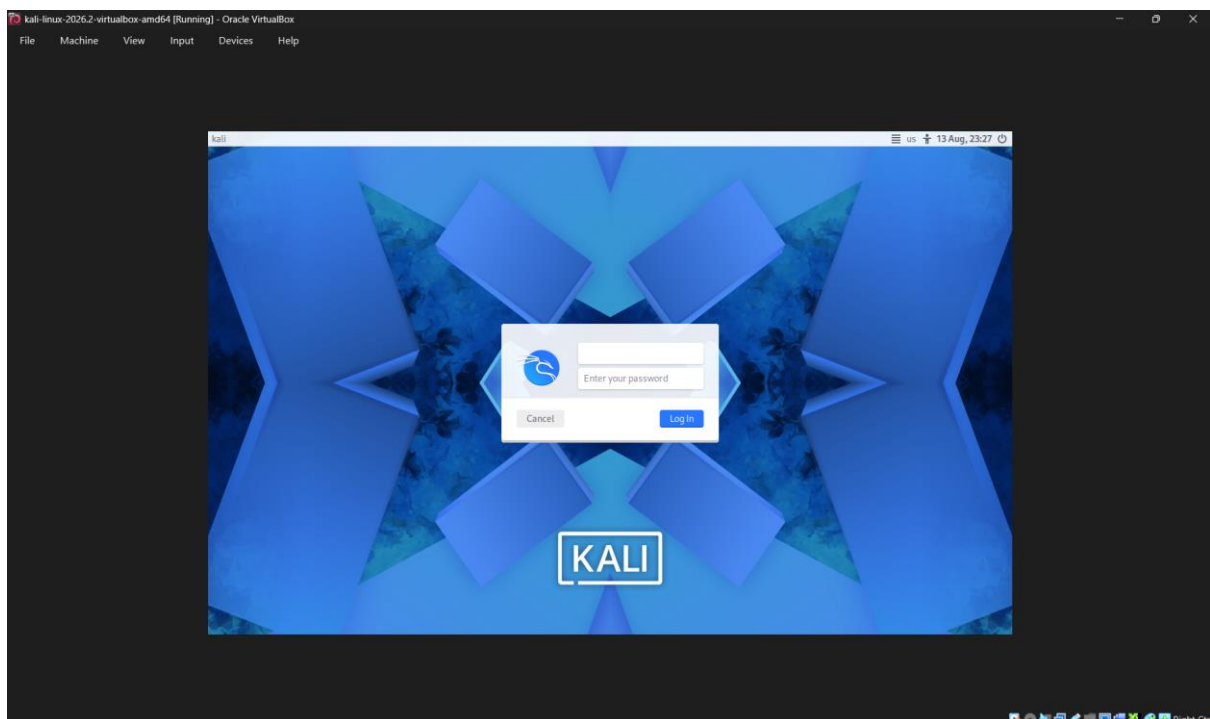
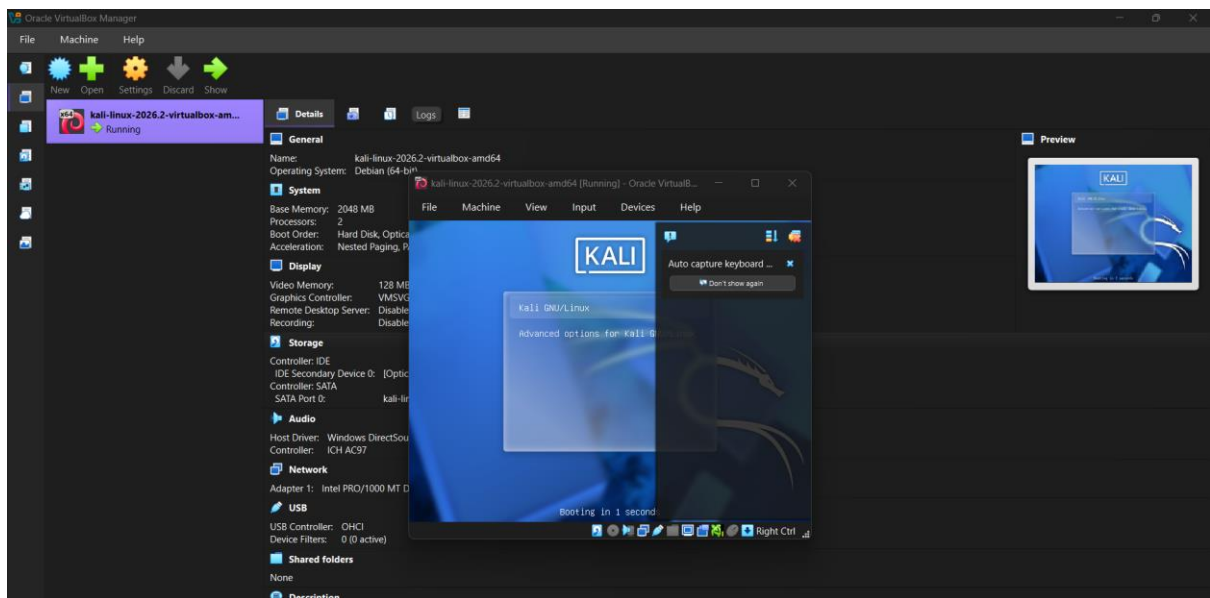
Step 5: Open Oracle VirtualBox.



Step 6: Click "Add" or "Open" and select the extracted Kali Linux VM file.



Step 7: Click "Start" to launch the Kali Linux virtual machine.



Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.

- **Step 8:** Open terminal inside Kali, create a simple C program file, compile it with gcc, and run it to make sure the environment works properly.

3. Working with Linux Commands

Terminal Basics

```
Terminal  Shell  Edit  View  Window  Help
[manalithakar@Manalis-MacBook-Air ~ % pwd
/Users/manalithakar
[manalithakar@Manalis-MacBook-Air ~ % mkdir "exp 1"
mkdir: exp 1: File exists
[manalithakar@Manalis-MacBook-Air ~ % cd "exp 1"
[manalithakar@Manalis-MacBook-Air exp 1 % pwd
/Users/manalithakar/exp 1
[manalithakar@Manalis-MacBook-Air exp 1 % mkdir temporary
[manalithakar@Manalis-MacBook-Air exp 1 % ls
temporary
[manalithakar@Manalis-MacBook-Air exp 1 % rmdir temporary
manalithakar@Manalis-MacBook-Air exp 1 % clear
```

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```
[manalithakar@Manalis-MacBook-Air ~ % history
33 dir
34 ls
35 clear
36 man ls
37 pwd
38 mkdir "exp 1"
39 cd "exp 1"
40 pwd
41 mkdir temporary
42 ls
43 rmdir temporary
44 clear
45 histroy
46 man ls
47 whoami
48 vi file.txt
[manalithakar@Manalis-MacBook-Air ~ % man
[manalithakar@Manalis-MacBook-Air ~ % whoami
manalithakar
manalithakar@Manalis-MacBook-Air ~ %
```

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- pwd: Shows the current working folder path.
- mkdir: Makes a new directory.
- rmdir: Deletes an empty folder.
- dir: Lists folder contents horizontally.
- ls: Lists files and folders in the current location.
- cd: Changes the current folder.
- cd ..: Goes back to the parent folder.
- clear: Clears the terminal screen.
- history: Shows previously typed commands.
- man: Opens the manual for a command.
- whoami: Displays the logged-in user name.

Files

- vi: Opens a text editor to create or edit files.
- file: Shows file type details.
- rm: Deletes a file.
- cp: Copies files or folders to another folder.
- mv: Moves or renames files and folders.
- cat: Prints file contents on screen.
- more: Displays file content page by page.
- less: Views file content with scroll support.
- head: Displays the first few lines of a file.
- tail: Displays the last few lines of a file.
- chmod: Changes file read/write permissions.

Redirection

- >: Overwrites output into a file.
- >>: Appends output to the end of an existing file.
- <: Takes input from a file instead of the keyboard.